

PLAYDB – Video Game Database & Marketplace System

DELIVERABLE 1(C)

Main Deliverable

Course: Software Engineering

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Team Name: Respawn Studio

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1. Introduction

PlayDB is a web-based video game tracking and marketplace system designed to provide gamers with a centralized platform to manage their gaming activities. The system allows users to register, track their playtime, maintain personalized game lists, and interact with other players. In addition to tracking features, PlayDB includes a localized marketplace where users in Pakistan can buy and sell gaming equipment and digital gaming products. The system also aims to integrate AI-based recommendations to help users discover new games based on their preferences.

2. Project Vision

The long-term vision of PlayDB is to become a comprehensive gaming ecosystem that combines social tracking, personalized game management, and secure local marketplace functionality within a single platform. The system aims to support the growing gaming community in Pakistan by providing a reliable and user-friendly solution for tracking gaming activity and facilitating trusted buying and selling. In the future, PlayDB can be expanded to include advanced analytics, enhanced AI recommendations, and broader regional support.

3. Intended Use of the System

Stakeholders

The primary stakeholders of the PlayDB system include:

- Gamers (end users)
- Buyers and sellers of gaming products
- System administrators
- Development team

Needs

- **Gamers** need a platform to track their playtime, create personalized lists, and receive game recommendations.
- **Buyers** require a secure and reliable marketplace to purchase gaming equipment and digital products locally.
- **Sellers** need an easy-to-use system to list products and connect with potential buyers.
- **System administrators** require tools to manage users, monitor listings, and maintain system security.

- The PlayDB system is intended to meet these needs by providing an integrated, efficient, and user-friendly platform.

4. Features / Overall Functionality

The PlayDB system will provide an integrated platform combining gaming activity tracking, social interaction, and marketplace functionality. The major features of the system are described below:

4.1 User Management

- User registration and secure login
- Profile creation and profile management
- Session handling and logout functionality

This feature ensures secure access to the system and personalized user experience.

4.2 Game Tracking System

- Add games to personal library
- Track playtime for each game
- Maintain lists such as “Currently Playing,” “Completed,” and “Favorites”
- Create personalized top 10 lists (e.g., favorite soundtracks, characters, or genres)

This functionality helps users organize and monitor their gaming activities efficiently.

4.3 Social Features

- View friends’ gaming activity
- Compare playtime and favorite games
- Share gaming preferences

These features promote community engagement and interaction among users.

4.4 Marketplace Module (Pakistan Only)

- List gaming equipment for sale
- Buy and sell physical gaming products locally
- Post digital products such as games or battle passes
- Contact sellers through the platform

This module supports local gamers by enabling secure and convenient transactions within Pakistan.

4.5 AI Recommendation System

- Analyze user preferences and play history
- Suggest new games based on interests
- Provide personalized recommendations

The AI chatbot enhances user experience by helping players discover new games tailored to their tastes.

4.6 Administrative Controls

- Manage users and listings
- Monitor inappropriate content
- Maintain system security and performance

Overall, PlayDB will function as a centralized gaming platform that simplifies tracking, enhances social interaction, and enables safe local trading.

5. User Stories

User Story 1 – User Registration

As a new user

I want to create an account

So that I can access the PlayDB system and track my gaming activity.

Sub-Tasks:

- Design registration form (UI)
- Validate user inputs (email, password, username)
- Store user data in database
- Display success or error message

User Story 2 – User Login

As a registered user

I want to log in to my account

So that I can access my personalized dashboard.

Sub-Tasks:

- Design login form
- Validate credentials against database
- Create user session
- Redirect to dashboard on success
- Display error message on invalid login

User Story 3 – View Profile

As a registered user

I want to view my profile information **So that** I can see my account details.

Sub-Tasks:

- Create profile page UI
- Retrieve user data from database
- Display username, email, and basic information

6. Structured Specifications

Structured Specification – User Registration

1. Input:

- Username
- Email address
- Password

2. Processing:

- Validate that all fields are filled.
- Check if email format is valid.
- Verify password meets minimum security requirements.
- Check database to ensure user does not already exist.
- Encrypt password.
- Store user data in the database.

3. Output:

- Success message: “Registration successful.” OR
- Error message indicating invalid input or existing user.

4. Preconditions:

- User is not already registered.
- Database connection is active.

5. Postconditions:

- New user account is successfully created and stored in the system.

Structured Specification – User Login

1. Input:

- Email/Username
- Password

2. Processing:

- Validate that fields are not empty.
- Retrieve user record from database.
- Compare entered password with encrypted password.
- If valid, create user session.

3. Output:

- Redirect to user dashboard on successful login.
OR
- Error message: “Invalid credentials.”

4. Preconditions:

- User account already exists.
- Database is accessible.

5. Postconditions:

- User session is created and user gains access to the system.

Structured Specification – View Profile

1. Input:

- Profile page request from logged-in user.

2. Processing:

- Retrieve user data from database.
- Fetch username, email, and other stored information.
- Display data on profile page.

3. Output:

- User profile information displayed on screen.

4. Preconditions:

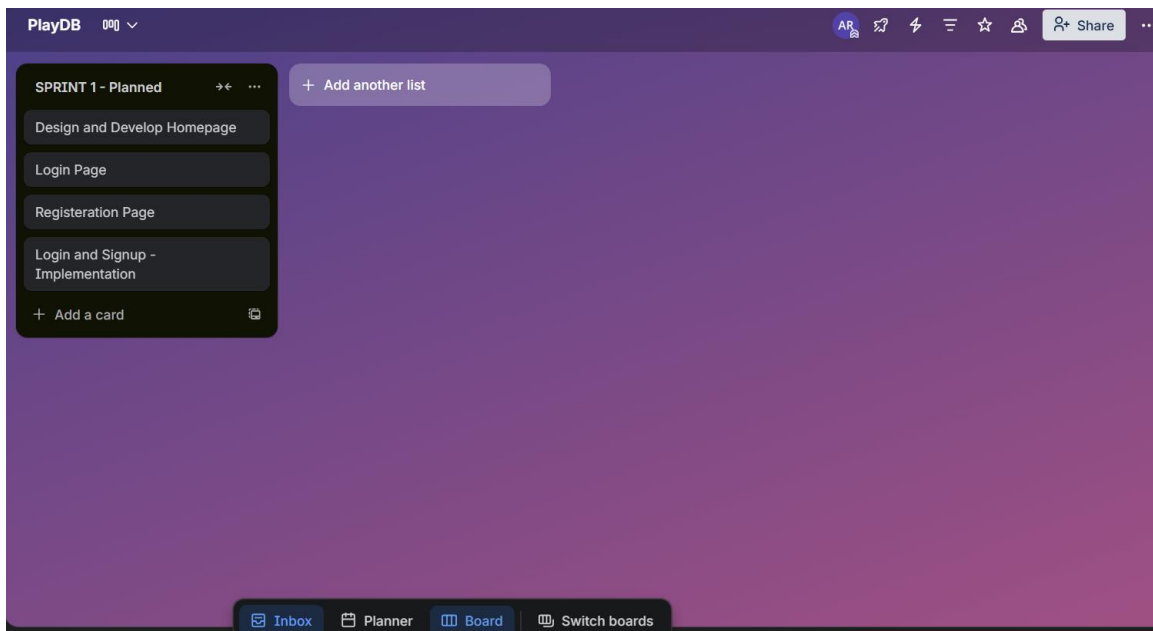
- User must be logged in.
- User data must exist in database.

5. Postconditions:

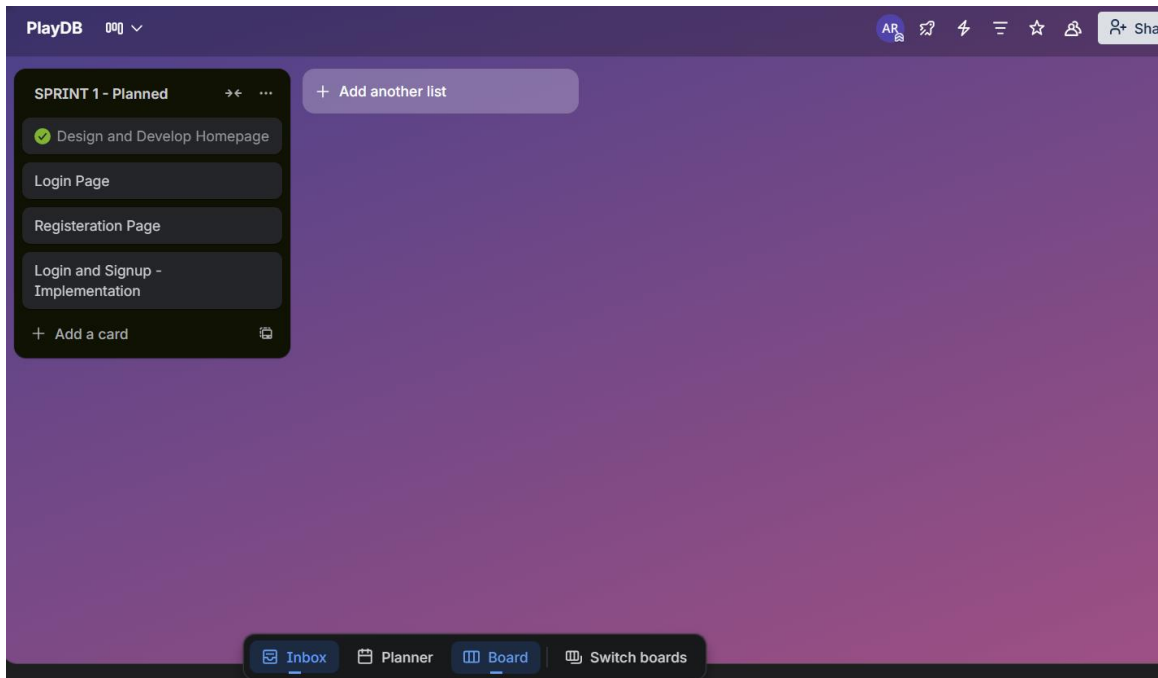
- Profile information is visible to the user.

7. Scrum Board

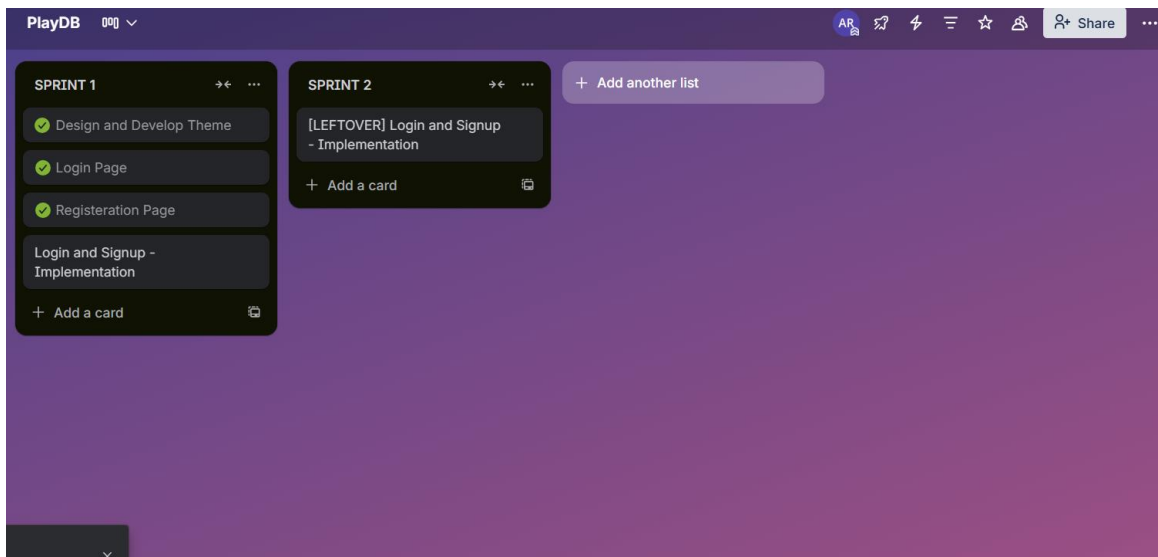
Initial Backlog



Mid Sprint



End Sprint



8. Non-Functional Requirements (NFR)

Non-functional requirements define the quality attributes and operational constraints of the PlayDB system.

8.1 Performance Requirements

- The system should respond to user actions within 2–3 seconds under normal load conditions.
- Login and registration processes should complete within 5 seconds.
- The system should efficiently handle multiple concurrent users without significant performance degradation.

8.2 Security Requirements

- User passwords must be encrypted before being stored in the database.
- Only authenticated users should access protected pages.
- User session management must prevent unauthorized access.
- Marketplace transactions must be restricted to registered users.

8.3 Usability Requirements

- The system should have a user-friendly and intuitive interface.
- Navigation should be simple and clearly structured.
- The system should require minimal technical knowledge to operate.

8.4 Reliability Requirements

- The system should operate without crashes during normal usage.
- Data entered by users should be saved accurately without loss.
- The system should maintain data integrity in case of unexpected shutdowns.

8.5 Availability Requirements

- The system should be accessible 24/7, except during scheduled maintenance.
- Maintenance activities should be communicated in advance.

8.6 Scalability Requirements

- The system should be designed to support increasing numbers of users in the future.
- Database structure should allow expansion without major redesign.

8.7 Maintainability Requirements

- The system should follow modular design principles for easier updates.
- Code should be properly documented to support future development and debugging.

9. Iteration 1 Implementation Screenshots



PLAYDB

GUIDED BY TASTE



RECBOT

Find your next game using an AI chatbot

RANDOM GAME

Get recommended a random videogame

WHAT'S TRENDING

See popular games right now

TOP RATED

Hall of fame titles

REVIEWS

Read expert critiques

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Compete for the top spot

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We are here to help





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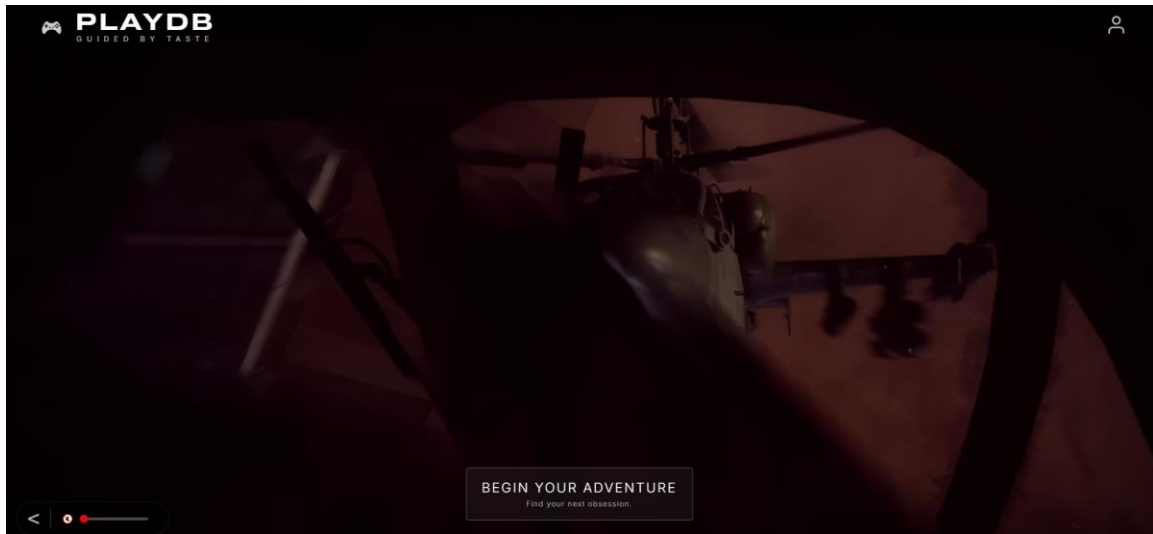
LEADERBOARDS

Compete for the top spot

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10. Work Division

All team members collaboratively decided the main features and overall structure of the PlayDB system. The initial layout and design ideas for the homepage, login, and registration screens were discussed and finalized collectively.

Abdullah Razzaq (Lead UI Developer) primarily implemented the user interface design with help from both other team members. All team members contributed equally to the preparation and documentation of the project deliverables.